Actual chart units and the algorithm obligation

November9,2026. Source `7bb276623`. Unsigned producer evidence throughout [pcc:unverified].

New actual-object result. The rational denominator and earlier scalar normalizer evaluate to units at every interpolation value in every commutative coefficient target where the image of two is nilpotent. This includes every $\mathbb{Z}/(2^p)$ value and permits further nilpotents/zero divisors. A constructed normalized finite entry differs from the old scalar-square entry by one actual common unit, preserving every scalar divisibility threshold.

Chart proof. Write $\mathcal{O}$ for the actual rational integral ring and $\alpha = 2^{a+1}$. Actual nonzero differences satisfy $|m-j| < \alpha$, hence $v_2(m-j) \le a$. Therefore $\alpha/(m-j)$ is even in $\mathcal{O}$; the zero equal-index ratio is even too. In the already constructed puncture chart,

$$B_s(t_m + \alpha z) = \rho_m \begin{cases} 1 & s = m \\ \alpha z/(m-s) & s \neq m \end{cases} \prod_{j \neq m,s} (1 + \alpha z/(m-j)).$$

Under any coefficient map killing two, this polynomial is constant ρ_m when $s = m$ and zero otherwise. Consequently the actual polynomials satisfy

$$D = \sum_s B_s \equiv \rho_m \pmod 2, \qquad R = \sum_s \rho_s^{-2} B_s \equiv \rho_m^{-1} \pmod 2.$$

The old normalizer's actual rational formula is connected to this integral polynomial. Its definition is exposed for the module API, without changing its value or any previous proof body.

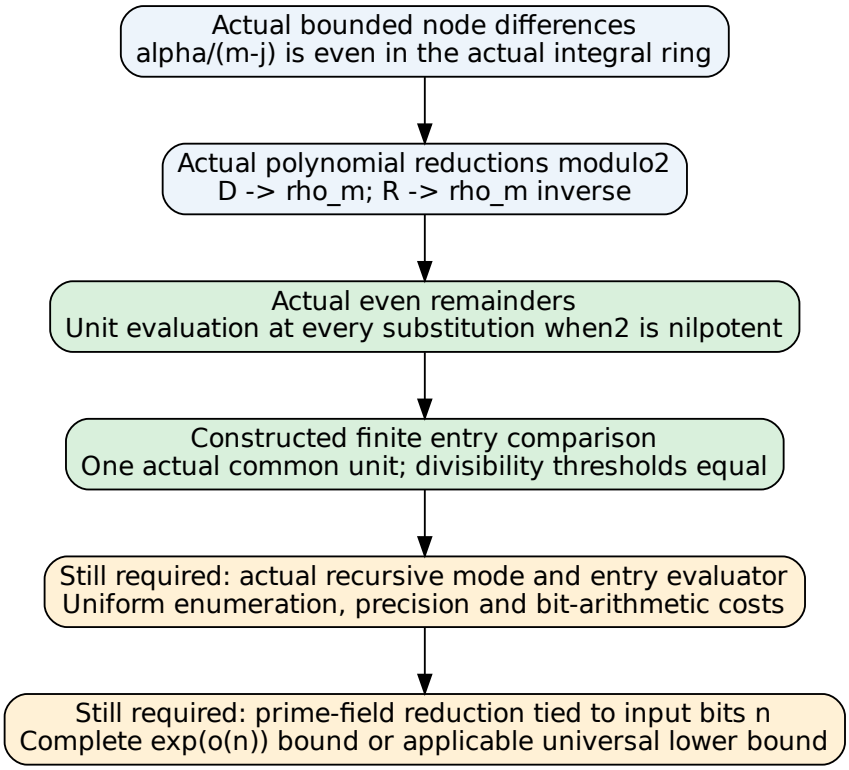
Units after substitution. The actual residue kernel proves $D - C(\rho_m)$ and $R - C(\rho_m^{-1})$ divisible by $C(2)$. Evaluation under $\pi : \mathcal{O} \to A$ gives a unit plus $\pi(2)$ times a value. If $\pi(2)$ is nilpotent, the perturbation is nilpotent, so the sum is a unit for every $z \in A$. No assumed unit for an unspecified normalizer is used.

Finite comparison and verification

Let n_s be the actual base-chart evaluation and d the unit produced for D. Define $U_s = d^{-1}n_s^2$. If r is the actual old-normalizer evaluation, then $c = rd$ is a unit and

$$cU_s = rn_s^2, \qquad e \mid rn_s^2 \iff e \mid U_s.$$

This does not prove a minimal Smith/Newton factorization, an optimal mode count or a full recursive cost. Historical panel rows remain unchanged and are not relabelled as new rational-kernel measurements.



Nine unsigned PCC bindings (prefix `PCC-20261109-`): even ratio777347, denominator reduction9c4c4f, old-normalizer evaluation65b9f5, normalizer reduction53504f, denominator units485e5c, normalizer units7cea35, modular comparison units c9a342, finite entry identity51ff85, divisibility ff3e71. Canonical hashes and successful/failed logs are retained under `lean/`. Actual source audit and fresh import-inclusive status are appended to the companion report after completion. No new native census was run.

Goal assessment and exact remaining dependency

The requested faithful Lean proof of existence or nonexistence of a novel classical sub-exponential algorithm for general prime-field ECDLP is still absent. The unresolved dependency is mathematical, not machine access.

Existence route. Actual clearing, finite recovery and this unit bridge provide algebraic components. A complete program still needs an actual all-level mode/child-entry evaluator, packing and input construction, success/recovery proofs and every construction, precision, failed-attempt and bit-arithmetic cost. Its reduction must connect network/circuit size s to the ECDLP input length n and establish total $T(n) = \exp(o(n))$. Identifying s with n without a proved reduction is insufficient. Supplied tensors or coefficient buffers do not establish their own efficient construction.

Representation alternative. The scaled accumulator has a proved scaled target congruence and valid cross-precision cancellation. Native canonical extraction, coefficient delivery, retained-state and recomputation costs remain separate. The closed native census supplies finite correctness, not asymptotic cost evidence.

Origin and jet alternatives. A physical-coordinate transformation need not act as a Boolean gauge; the prior finite counterexample remains. A correction $W+C(T)H$ with $C(0) = 0$ retains $W(0)$, although higher jets remain adjustable. Neither fact rules out a different target-preserving channel or a future all-level rank construction.

Nonexistence route. These family-specific facts are not a lower bound for every non-generic algorithm. An algorithmic upper-bound envelope cannot supply that lower bound. No applicable unrestricted lower-bound theorem is established here. Assuming the desired runtime or correctness in a wrapper would not constitute the requested proof.

Sources. Actual formal chart/residue modules and earlier reports; pinned OpenAI Oct25,2026 manuscript D.4.3–D.4.7/D.9/D.10; primary Mathlib PadicNorm and Nilpotent.Basic. KB unavailable. No priority or exhaustive-search claim. Exact next obligations are in PLAN.md and the current goal-status note.